Homeostasis

Exam code: 0610 & 0970

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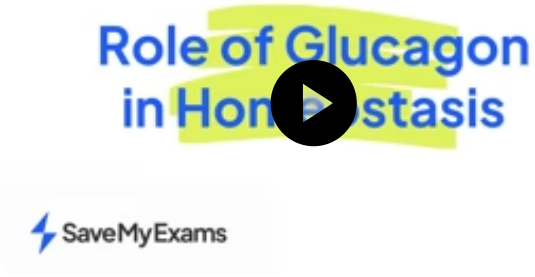

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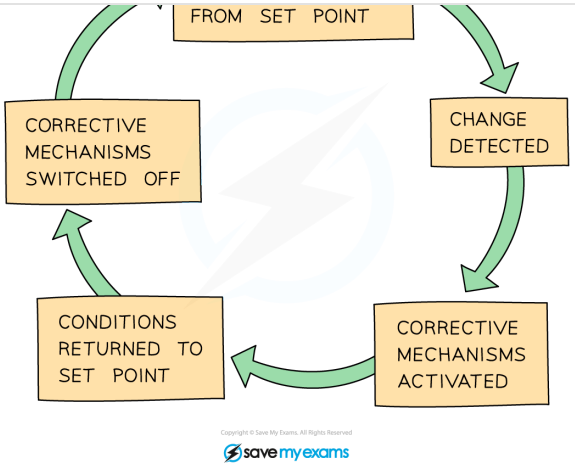
The Concept of Negative Feedback: Extended

- Negative feedback occurs when conditions change from the ideal or **set point** and returns conditions to this set point
- It works in the following way:
  - if the level of something **rises**, control systems are switched on to **reduce it** again
  - if the level of something **falls**, control systems are switched on to **raise it** again
- Negative feedback mechanisms are usually a continuous cycle of bringing levels down and then bringing them back up so that overall, they stay within a **narrow range** of what is considered '**normal**'

Help me understand

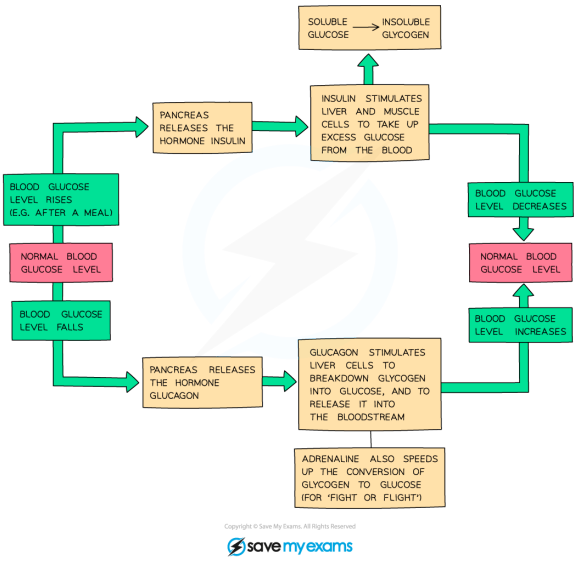
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



The negative feedback cycle

Blood Glucose Control: Extended

- Blood glucose levels are controlled by a negative feedback mechanism involving the production of two **hormones** - **insulin** and **glucagon**
- Both hormones which control blood glucose concentration are made in the **pancreas**
- **Insulin** is produced when **blood glucose rises** and **stimulates liver and muscle cells to convert excess glucose into glycogen to be stored**
- **Glucagon** is produced when **blood glucose falls** and **stimulates liver and muscle cells to convert stored glycogen into glucose to be released into the blood**



Negative feedback regulation of blood glucose levels

Examiner Tips and Tricks

The terms glucagon and glycogen are very often mixed up by students as they sound similar. Remember:

- Glucagon is the **hormone**
- Glycogen is the polysaccharide **glucose is stored as**

Learn the differences between the spellings and what each one does so you do not get confused in the exam!

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Type 1 Diabetes: Extended

- **Type 1 diabetes** is a condition where the blood glucose levels are not able to be regulated as the **insulin-secreting cells in the pancreas are not able to produce insulin**
- This means that blood glucose levels are often **far too high**
- It can be treated by **injecting insulin**
- The extra insulin causes the liver to convert **glucose into glycogen**, which **reduces** the blood glucose level
- **Symptoms** of diabetes include extreme thirst, weakness or tiredness, blurred vision, weight loss and loss of consciousness in extreme cases
- People with Type 1 diabetes have to **monitor** their blood glucose levels throughout the day as their levels of **physical activity** and their **diet** affect the amount of insulin needed
- They can help to **control** their blood glucose level by being careful with their **diet** - eating foods that will not cause large increases in blood glucose level, and by **exercising**, which can lower blood glucose levels due to increased respiration in the muscles

Examiner Tips and Tricks

lifestyle factors.

You only need to know the symptoms and treatment of Type 1

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